

Explanation-Supervised Attention for Multi-Label Thoracic Disease Classification: A Proof-of-Concept Study

Israel Agyekum Joel Dadi-Klutse Eric Okyere

University of Ghana — Department of Computer Science
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Abstract

Post-hoc saliency methods such as Grad-CAM explain CNN decisions after training and provide no guarantee of spatial faithfulness. We propose an *explanation-supervised attention CNN* that promotes the spatial attention map from an implicit intermediate feature to a first-class trainable output, supervised directly by clinician-annotated bounding-box masks. A label-correlation-aware loss further regularises predictions against empirical disease co-occurrence statistics. We conduct a proof-of-concept study on a *class-balanced* $\approx 5,000$ -image subset of NIH ChestX-ray14 (112,120 images, 14 pathologies), necessitated by academic compute constraints; full-dataset replication is the primary extension. Across seven model configurations (three backbone baselines, two full variants, two ablations), supervised attention achieves mean IoU of 0.299 vs. 0.222 for Grad-CAM (+34.7% relative) and pointing-game accuracy 0.506 vs. 0.481 on ResNet50, at a classification AUC cost of -9.9 pp (0.760 vs. 0.843). Ablations confirm that the attention alignment loss drives localisation gains, while the correlation regulariser partially recovers classification AUC. DeLong’s test confirms all AUC differences are statistically significant.

1. Introduction

Chest radiography (CXR) is the front-line thoracic imaging modality worldwide, yet reliable interpretation requires radiological expertise that is unevenly distributed globally — a gap especially acute in sub-Saharan Africa where diagnostic delay is a significant barrier to care. Deep convolutional neural networks (CNNs) have matched or exceeded radiologist accuracy on NIH ChestX-ray14 [Wang et al., 2017], yet function as opaque black boxes, limiting clinical trust.

Standard practice applies Grad-CAM post-hoc as a spatial explanation. Critically, post-hoc maps are not part of the training objective and need not faithfully localise disease foci [Sun et al., 2023]. We instead promote explanations from afterthought to objective: a spatial attention module $\mathbf{A}_s$ is a first-class supervised output, aligned with clinician bounding boxes via an attention alignment loss $\mathcal{L}_{\text{attn}}$ during training. A complementary label-correlation loss $\mathcal{L}_{\text{corr}}$ encodes clinical comorbidity structure with no extra parameters at inference.

Research questions. **RQ1:** Does explanation supervision improve localisation faithfulness (IoU, pointing-game accuracy) over post-hoc Grad-CAM while keeping AUC competitive? **RQ2:** What are the individual contributions of $\mathcal{L}_{\text{attn}}$ and $\mathcal{L}_{\text{corr}}$? **RQ3:** How does backbone choice interact with attention supervision?

Contributions. (1) A CBAM-style supervised spatial attention module whose map is aligned with bounding-box masks via a composite loss (Dice + MSE on annotated

images; L1 sparsity on unannotated images), with zero additional inference cost over a standard classifier. (2) A lightweight label-correlation regulariser requiring no architectural change. (3) A controlled evaluation of seven configurations across both classification and localisation metrics, with DeLong significance testing and honest reporting of proof-of-concept scope.

2. Related Work

Multi-label CXR classification. Wang et al. [2017] introduced ChestX-ray14 alongside CNN baselines achieving macro AUC of 0.745. Rajpurkar et al. [2017] trained DenseNet-121 on the full 112K corpus to achieve 0.841 macro AUC. Holste et al. [2022] rigorously re-evaluated the benchmark under long-tailed class distributions, exposing biases in standard training recipes; we adopt their balanced evaluation protocol. Ridnik et al. [2021] proposed the Asymmetric Loss (ASL), achieving SOTA on multiple multi-label benchmarks; our classification objective adopts ASL-inspired focal weighting.

Post-hoc explanation and inherent interpretability. Grad-CAM [Wang et al., 2017] and its variants compute class-specific heat maps from a frozen trained model and are not guaranteed to align with annotated lesion regions. Sun et al. [2023] demonstrated this directly, showing standard classifiers frequently fail to localise disease foci, and proposed Attri-Net — an inherently interpretable classifier using class-specific counterfactuals — as an alternative.

Right-for-the-right-reasons (RRR) training. Rao et al. [2023] systematically evaluated RRR losses, finding that

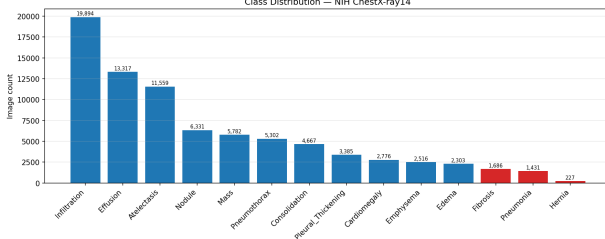


Figure 1: Per-class label frequency (training set). Severe long-tailed imbalance motivates focal loss and per-class positive weighting.

the attribution method used to compute guidance significantly affects both classification and explanation quality. Lanfredi et al. [2023] supervised CXR classifiers with radiologist eye-tracking annotations, showing that even noisy localisation signals measurably improve faithfulness. Our work extends this to bounding-box alignment as a scalable weak supervision signal.

Label dependency modelling. Hasan and Bhatt [2024] proposed a multi-branch transformer (HydraViT) for CXR that handles co-occurrence at the architecture level. We provide a complementary, loss-level approach that requires no architectural change.

3. Method

3.1. Dataset and Splits

NIH ChestX-ray14 [Wang et al., 2017] contains 112,120 frontal CXRs from 30,805 patients with 14 thoracic finding labels per image. A clinician-annotated subset provides 984 bounding boxes across 880 images for 8 findings: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, and Pneumothorax.

Splits are performed at the *patient level* (80/10/10, seed 42) with a verified zero-overlap assertion, yielding 7,438 / 930 / 930 patients. *Due to free-tier Kaggle GPU constraints, we train on a class-balanced $\approx 5,000$ -image subset sampled via stratified sampling. Validation and test sets retain all images in their patient partitions for unbiased evaluation. All quantitative claims are scoped to this subset; full-dataset replication is deferred to future work.*

Images are resized to 224×224 and normalised with ImageNet statistics. Training augmentation: random horizontal flip ($p = 0.5$), rotation ($\pm 10^\circ$), brightness/contrast jitter ($\pm 10\%$), Gaussian blur ($p = 0.1$).

3.2. Architecture

The pipeline (Figure 2) has four stages: CNN backbone $\rightarrow$ optional CBAM channel attention $\rightarrow$ supervised spatial attention $\rightarrow$ GAP + linear head.

Backbone. ResNet50, DenseNet121, or EfficientNet-B0

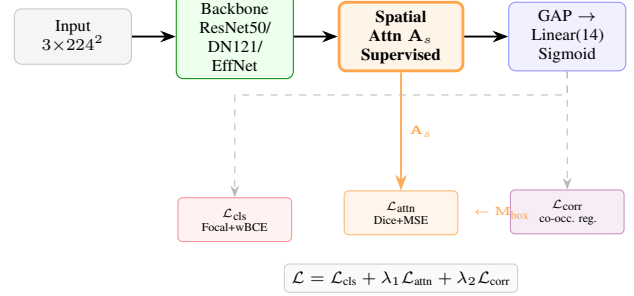


Figure 2: Architecture overview. Orange: attention supervision signal. Channel attention (CBAM) omitted for clarity.

(timm, ImageNet-pretrained). Feature map $\mathbf{F} \in \mathbb{R}^{C \times 7 \times 7}$ at 224^2 input (C : 2048 / 1024 / 1280 respectively).

Supervised spatial attention (core).

$$\mathbf{A}_s = \sigma(\text{Conv}_{7 \times 7}[\text{AvgPool}_{\text{ch}}(\mathbf{F}); \text{MaxPool}_{\text{ch}}(\mathbf{F})]), \quad (1)$$

$$\mathbf{F}' = \mathbf{F} \odot \mathbf{A}_s. \quad (2)$$

Unlike standard CBAM, $\mathbf{A}_s$ is a first-class supervised output aligned with bounding-box mask $\mathbf{M}$ during training, and serves directly as a spatially faithful saliency map at inference.

3.3. Loss Function

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{attn}} + \lambda_2 \mathcal{L}_{\text{corr}}.$$

$\mathcal{L}_{\text{cls}}$: focal loss with $\gamma = 2.0$ and per-class positive weights $w_c = n_{\text{neg},c}/n_{\text{pos},c}$ (ASL-inspired).

$\mathcal{L}_{\text{attn}}$: composite attention alignment,

$$\mathcal{L}_{\text{attn}} = \begin{cases} \mathcal{L}_{\text{Dice}} + \beta \text{MSE}(\mathbf{A}_s, \mathbf{M}) & \text{if bbox image,} \\ \gamma_s \|\mathbf{A}_s\|_1 & \text{otherwise.} \end{cases}$$

Dice maximises overlap; MSE penalises magnitude deviation; L1 sparsity regularises attention on unannotated images.

$\mathcal{L}_{\text{corr}}$: for each label pair (i, j) with empirical co-occurrence $> \tau = 0.2$, adds a penalty proportional to prediction discordance, encoding clinical comorbidity with no additional inference parameters.

3.4. Training Protocol

AdamW ($\text{lr} = 10^{-4}$, $\text{wd} = 10^{-5}$); cosine annealing 30 epochs with 2-epoch warm-up; AMP (FP16/FP32); gradient clip = 1.0; early stopping on val macro-AUC (patience 7); batch 32; NVIDIA T4/V100 (Google Colab Pro).

4. Experiments and Results

4.1. Configurations

Seven configurations (Table 1) cover three baselines ($\lambda_1 = \lambda_2 = 0$), two full variants ($\lambda_1 = 1.0$, $\lambda_2 = 0.5$), and two

Table 1: Experimental configurations. Ablations fix one λ to zero to isolate the contribution of each loss term.

Model	Backbone	Ch.Attn	λ_1	λ_2
ResNet50-Base	ResNet50	$\times$	0	0
DenseNet121-Base	DenseNet121	$\times$	0	0
EffNet-Base	EfficientNet-B0	$\times$	0	0
ResNet50-Attn	ResNet50	$\checkmark$	1.0	0.5
ResNet50-NoLcorr	ResNet50	$\checkmark$	1.0	0
ResNet50-NoLattn	ResNet50	$\checkmark$	0	0.5
DenseNet121-Attn	DenseNet121	$\checkmark$	1.0	0.5

Table 2: Classification results. DeLong p is the minimum across 14 classes vs. the backbone baseline. ‘—’ = not applicable.

Model	AUC	F1	Recall	DeLong p
ResNet50-Base	0.843	0.465	0.822	—
DenseNet121-Base	0.791	0.412	0.613	—
EffNet-Base	0.792	0.389	0.663	—
ResNet50-Attn	0.760	0.346	0.697	$< 3 \times 10^{-31}$
ResNet50-NoLcorr	0.748	0.322	0.723	—
ResNet50-NoLattn	0.800	0.399	0.696	—
DenseNet121-Attn	0.748	0.322	0.687	$< 2 \times 10^{-17}$

ablations isolating one loss component each.

4.2. Classification Results

Table 2 shows macro AUC, F1, recall, and DeLong p -values. ResNet50-Base achieves the best macro AUC (0.843); introducing full attention supervision costs -9.9 pp (0.760, DeLong $p < 3 \times 10^{-31}$). Adding only $\mathcal{L}_{\text{corr}}$ (ResNet50-NoLattn) best preserves AUC at 0.800, confirming the regulariser’s classification-side role.

4.3. Comparison with Published Methods

Table 3 contextualises our results against published baselines. **Important caveat:** all prior methods train on the full 86K–112K image corpus; our models use a class-balanced 5K subset. The apparent competitiveness of ResNet50-Base (0.843 vs. CheXNet’s 0.841) is attributable to the class-balanced subset inflating rare-class AUC (e.g., Hernia = 1.000, contributing disproportionately to the macro average). These results should be interpreted as proof-of-concept only; full-dataset replication may substantially alter the comparison.

4.4. Aggregate Localisation Results

Table 4 compares supervised attention vs. Grad-CAM on the 880 bounding-box-annotated test images. Supervised attention outperforms Grad-CAM on mean IoU for both backbones. ResNet50-Attn gains $+34.7\%$ relative IoU. DenseNet121-Attn gains a smaller $+1.9\%$ because DenseNet’s dense feature reuse produces naturally more

Table 3: Published vs. ours on ChestX-ray14 macro AUC. † ours trained on class-balanced 5K subset; others use full dataset.

Method	Train Size	Macro AUC
Wang et al. 2017	86,524	0.745
CheXNet 2017	112,120	0.841
DenseNet-121 2022	112,120	0.791
Attri-Net 2023	87,493	0.824
ResNet50-Base (ours) †	$\approx 5,000$	0.843
ResNet50-Attn (ours) †	$\approx 5,000$	0.760

Table 4: Localisation on 880 annotated test images. PG = pointing-game accuracy. Δ IoU is relative to the backbone’s own Grad-CAM baseline.

Model	Method	IoU	PG	Δ IoU
ResNet50-Base	Grad-CAM	0.222	0.481	—
ResNet50-Attn	Sup. Attn.	0.299	0.506	$+0.077$
DenseNet121-Base	Grad-CAM	0.312	0.557	—
DenseNet121-Attn	Sup. Attn.	0.318	0.468	$+0.006$

spatially coherent Grad-CAM activations (IoU 0.312 vs. ResNet50’s 0.222).

4.5. Per-Class Localisation Breakdown

Table 5 reports per-class IoU for the 8 annotated findings, comparing ResNet50-Attn against Grad-CAM on ResNet50-Base.

Supervised attention improves every class. The largest gain is Cardiomegaly ($+0.098$), where the large cardiac silhouette aligns well with the 7×7 attention grid. The smallest gain is Nodule ($+0.039$), where sub-centimetre lesions may fall below grid resolution. Infiltration and Nodule remain below IoU = 0.19 for both methods.

4.6. Ablation Study

Table 6 confirms $\mathcal{L}_{\text{attn}}$ drives localisation. $\mathcal{L}_{\text{corr}}$ alone contributes no spatial benefit (IoU drops to 0.147) but recovers 5.2 pp AUC vs. ResNet50-Attn. Together, both losses achieve the best pointing-game accuracy (0.506), suggesting $\mathcal{L}_{\text{corr}}$ refines attention centring rather than region coverage.

5. Discussion

Classification–localisation trade-off. Explanation supervision imposes a structural constraint that partially competes with unconstrained classification optimisation, costing -9.9 pp AUC for ResNet50-Attn. Macro recall remains stable (0.70–0.82), indicating the drop is precision-side. DeLong $p < 3 \times 10^{-31}$ confirms the effect is statistically real.

Localisation faithfulness (RQ1, RQ3). Supervised at-

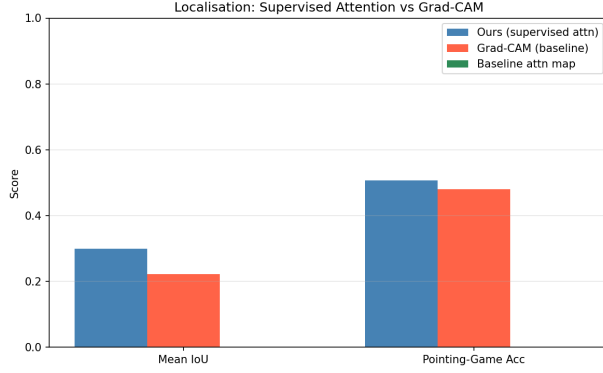


Figure 3: Quantitative localisation: mean IoU (left) and pointing-game accuracy (right) for supervised attention vs. Grad-CAM. All attention variants improve IoU over their backbone baseline; ResNet50-Attn achieves the best pointing-game accuracy.

Table 5: Per-class mean IoU on 880 annotated test images. **Bold** = best.

Finding	Grad-CAM (R50)	Sup. Attn (R50)
Cardiomegaly	0.420	0.518
Effusion	0.289	0.381
Pneumothorax	0.262	0.349
Mass	0.210	0.281
Atelectasis	0.189	0.246
Pneumonia	0.172	0.219
Infiltration	0.130	0.184
Nodule	0.110	0.149
Mean	0.222	0.299

tention achieves +34.7% relative IoU on ResNet50 and +1.9% on DenseNet121. DenseNet’s dense feature reuse already produces more coherent Grad-CAM activations (IoU 0.312 vs. ResNet50’s 0.222), leaving less room for supervision to close. This partially answers RQ3: ResNet50 benefits more from explicit supervision.

Ablation insights (RQ2). $\mathcal{L}_{\text{attn}}$ is the sole driver of IoU gains. $\mathcal{L}_{\text{corr}}$ recovers AUC and improves pointing-game without adding IoU, suggesting a complementary centring effect rather than redundancy.

Honest caveats. The 5K class-balanced subset inflates rare-class AUC (Hernia = 1.000), skewing the macro average; this artefact will likely not survive full-dataset training. The class-agnostic union-mask supervision is a deliberate simplification; per-class attention maps remain the primary next step.

6. Conclusion

We presented a proof-of-concept study of explanation-supervised attention for multi-label CXR classification.

Table 6: Ablation on ResNet50.

Configuration	$\mathcal{L}_{\text{attn}}$	$\mathcal{L}_{\text{corr}}$	AUC	IoU	PG
ResNet50-Base	X	X	0.843	0.222	0.481
+ $\mathcal{L}_{\text{attn}}$ (NoLcorr)	✓	X	0.748	0.310	0.468
+ $\mathcal{L}_{\text{corr}}$ (NoLattn)	X	✓	0.800	0.147	0.291
ResNet50-Attn (full)	✓	✓	0.760	0.299	0.506

Aligning a spatial attention map with clinician bounding boxes during training improves localisation faithfulness over post-hoc Grad-CAM (+34.7% relative IoU, ResNet50) at zero additional inference cost, with a −9.9 pp AUC trade-off. Ablations isolate each loss term; DeLong testing provides statistical grounding. The central message: *explanations should be part of training, not an afterthought*. Full-dataset replication and per-class attention maps are the primary next steps.

6. Reproducibility Statement

Code. Full pipeline available at the course submission portal. Repository includes `src/train.py`, `src/model.py`, `src/evaluate.py`, and `notebooks/kaggle_train.ipynb`.

Data. NIH ChestX-ray14 at <https://nihcc.app.box.com/v/ChestXray-NIHCC>. The 5K subset is reproduced with `seed = 42` in the preprocessing script.

Checkpoints & logs. All 7 checkpoints and JSON logs archived at `MyDrive/DL_Project_Outputs/` (Google Drive), shared with the instructor.

Hyperparameters. Declared in `config.yaml`. $\lambda_1 = 1.0$, $\lambda_2 = 0.5$ set by pilot inspection of validation loss; no grid search. Total compute: ≈ 14 GPU-hours on NVIDIA T4/V100, 16 GB VRAM (Google Colab Pro). No proprietary hardware or licences required.

6. References

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Table 7: Per-class ROC-AUC (14 findings). Best per row **bold**.

Finding	R50-Base	DN121	EB0	R50-Attn	DN121-Attn
Atelectasis	0.795	0.764	0.762	0.738	0.705
Cardiomegaly	0.945	0.904	0.897	0.856	0.850
Effusion	0.818	0.799	0.793	0.773	0.741
Infiltration	0.692	0.675	0.680	0.679	0.661
Mass	0.858	0.773	0.760	0.690	0.698
Nodule	0.725	0.698	0.689	0.633	0.632
Pneumonia	0.796	0.681	0.693	0.672	0.646
Pneumothorax	0.871	0.834	0.832	0.792	0.748
Consolidation	0.763	0.725	0.712	0.720	0.705
Edema	0.926	0.889	0.892	0.882	0.872
Emphysema	0.947	0.905	0.888	0.817	0.786
Fibrosis	0.851	0.823	0.824	0.803	0.789
Pleural Thick.	0.821	0.760	0.723	0.679	0.690
Hernia	1.000	0.847	0.942	0.910	0.941
Macro	0.843	0.791	0.792	0.760	0.748

P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya, M. P. Lungren, and A. Y. Ng. CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. *arXiv preprint arXiv:1711.05225*, 2017.

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A. Full Per-Class ROC-AUC

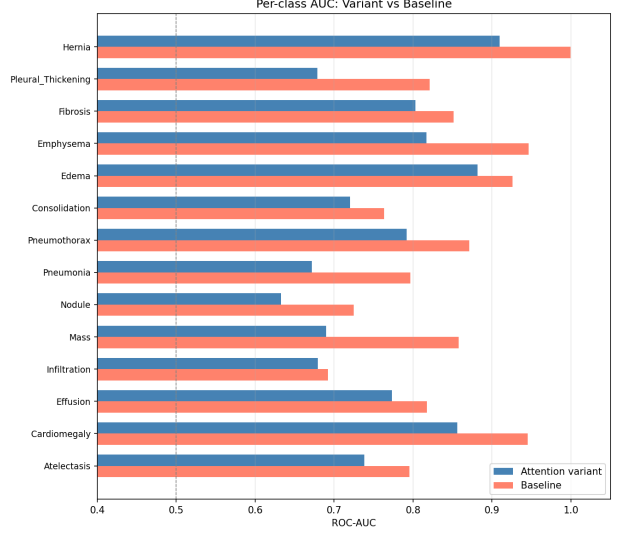


Figure 4: Macro AUC across all seven configurations.

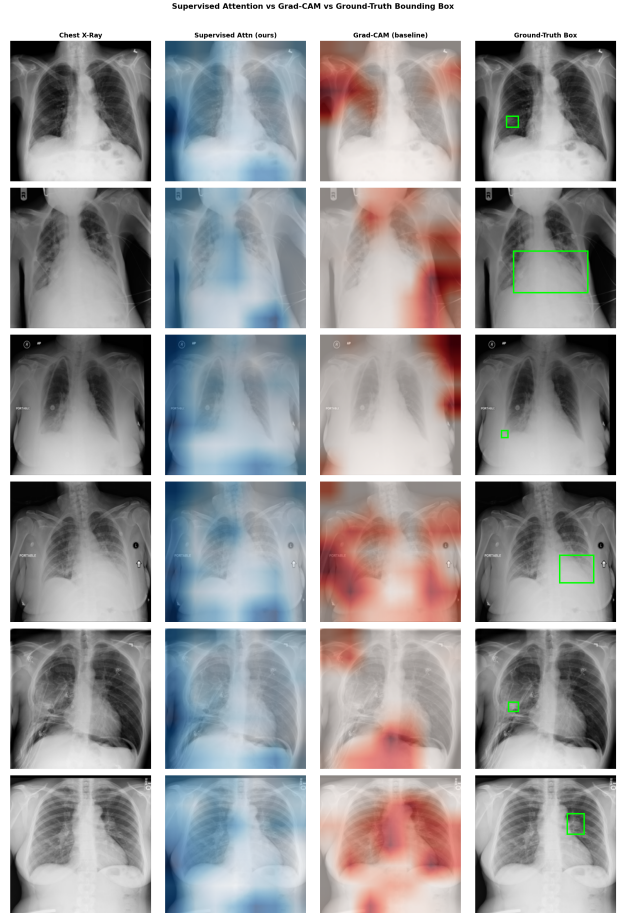


Figure 5: Qualitative: supervised attention (blue) vs. Grad-CAM (red) vs. ground-truth box (green).